```
akashh@akashhh: ~
akashh@akashhh:~$ hdfs namenode -format
23/06/29 22:03:30 INFO namenode.NameNode: STARTUP_MSG:
/************************************************************
STARTUP_MSG: Starting NameNode
STARTUP_MSG:   host = akashhh/127.0.1.1
STARTUP_MSG:   args = [-format]
STARTUP_MSG:   version = 2.7.3
STARTUP_MSG:   classpath = /home/akashh/Hadoop/etc/hadoop:/home/akashh/Hadoop/sh
are/hadoop/common/lib/jsr305-3.0.0.jar:/home/akashh/Hadoop/share/hadoop/common/l
ib/commons-io-2.4.jar:/home/akashh/Hadoop/share/hadoop/common/lib/commons-digest
er-1.8.jar:/home/akashh/Hadoop/share/hadoop/common/lib/jetty-6.1.26.jar:/home/ak
ashh/Hadoop/share/hadoop/common/lib/jersey-core-1.9.jar:/home/akashh/Hadoop/shar
e/hadoop/common/lib/commons-httpclient-3.1.jar:/home/akashh/Hadoop/share/hadoop/
common/lib/curator-client-2.7.1.jar:/home/akashh/Hadoop/share/hadoop/common/lib/
avro-1.7.4.jar:/home/akashh/Hadoop/share/hadoop/common/lib/api-asn1-api-1.0.0-M2
0.jar:/home/akashh/Hadoop/share/hadoop/common/lib/api-util-1.0.0-M20.jar:/home/a
kashh/Hadoop/share/hadoop/common/lib/jersey-json-1.9.jar:/home/akashh/Hadoop/sha
re/hadoop/common/lib/activation-1.1.jar:/home/akashh/Hadoop/share/hadoop/common/
lib/log4j-1.2.17.jar:/home/akashh/Hadoop/share/hadoop/common/lib/jackson-core-as
l-1.9.13.jar:/home/akashh/Hadoop/share/hadoop/common/lib/jersey-server-1.9.jar:/
home/akashh/Hadoop/share/hadoop/common/lib/asm-3.2.jar:/home/akashh/Hadoop/share
/hadoop/common/lib/xz-1.0.jar:/home/akashh/Hadoop/share/hadoop/common/lib/jsch-0
.1.42.jar:/home/akashh/Hadoop/share/hadoop/common/lib/hamcrest-core-1.3.jar:/hom
e/akashh/Hadoop/share/hadoop/common/lib/protobuf-java-2.5.0.jar:/home/akashh/Had
```

```
akashh@akashhh: ~
************************************************************/
akashh@akashhh:~$ start-all.sh
This script is Deprecated. Instead use start-dfs.sh and start-yarn.sh
Starting namenodes on [localhost]
localhost: starting namenode, logging to /home/akashh/Hadoop/logs/hadoop-akashh-
namenode-akashhh.out
localhost: starting datanode, logging to /home/akashh/Hadoop/logs/hadoop-akashh-
datanode-akashhh.out
Starting secondary namenodes [0.0.0.0]
0.0.0.0: starting secondarynamenode, logging to /home/akashh/Hadoop/logs/hadoop-
akashh-secondarynamenode-akashhh.out
starting yarn daemons
starting resourcemanager, logging to /home/akashh/Hadoop/logs/yarn-akashh-resour
cemanager-akashhh.out
localhost: starting nodemanager, logging to /home/akashh/Hadoop/logs/yarn-akashh
-nodemanager-akashhh.out
akashh@akashhh:~$ jps
27361 Jps
27253 NodeManager
26470 NameNode
26807 SecondaryNameNode
26616 DataNode
26957 ResourceManager
akashh@akashhh:~$
```

```
akashh@akashhh: ~
drwxr-xr-x   - akashh supergroup          0 2023-06-29 22:10 /testing/testing2
akashh@akashhh:~$ hdfs dfs -mv  /testing/sample.txt /testing/testing2
akashh@akashhh:~$ hdfs dfs -ls -R /
drwxr-xr-x   - akashh supergroup          0 2023-06-29 22:12 /testing
drwxr-xr-x   - akashh supergroup          0 2023-06-29 22:12 /testing/testing2
-rw-r--r--   1 akashh supergroup         21 2023-06-29 22:12 /testing/testing2/s
ample.txt
akashh@akashhh:~$ hdfs dfs -cat /testing/testing2/sample.txt
Hello Sample Program
akashh@akashhh:~$ ls
Desktop  Documents  Hadoop  Pictures  sample.txt  Templates  Videos
dfsdata  Downloads  Music   Public    snap        tmpdata
akashh@akashhh:~$ rm sample.txt
akashh@akashhh:~$ ls
Desktop  Documents  Hadoop  Pictures  snap       tmpdata
dfsdata  Downloads  Music   Public    Templates  Videos
akashh@akashhh:~$ hdfs dfs -get /testing/testing2/sample.txt
akashh@akashhh:~$ ls
Desktop  Documents  Hadoop  Pictures  sample.txt  Templates  Videos
dfsdata  Downloads  Music   Public    snap        tmpdata
akashh@akashhh:~$ hdfs dfs -rm /testing/testing2/sample.txt
23/06/29 22:14:23 INFO fs.TrashPolicyDefault: Namenode trash configuration: Dele
tion interval = 0 minutes, Emptier interval = 0 minutes.
Deleted /testing/testing2/sample.txt
```

```
akashh@akashhh: ~
starting resourcemanager, logging to /home/akashh/Hadoop/logs/yarn-akashh-resour
cemanager-akashhh.out
localhost: starting nodemanager, logging to /home/akashh/Hadoop/logs/yarn-akashh
-nodemanager-akashhh.out
akashh@akashhh:~$ jps
27361 Jps
27253 NodeManager
26470 NameNode
26807 SecondaryNameNode
26616 DataNode
26957 ResourceManager
akashh@akashhh:~$ hdfs dfs -mkdir /testing
akashh@akashhh:~$ hdfs dfs -ls -R /
drwxr-xr-x   - akashh supergroup          0 2023-06-29 22:10 /testing
akashh@akashhh:~$ hdfs dfs -mkdir /testing2
akashh@akashhh:~$ hdfs dfs -ls -R /
drwxr-xr-x   - akashh supergroup          0 2023-06-29 22:10 /testing
drwxr-xr-x   - akashh supergroup          0 2023-06-29 22:10 /testing2
akashh@akashhh:~$ hdfs dfs -mv  /testing2 /testing
akashh@akashhh:~$ hdfs dfs -ls -R /
drwxr-xr-x   - akashh supergroup          0 2023-06-29 22:11 /testing
drwxr-xr-x   - akashh supergroup          0 2023-06-29 22:10 /testing/testing2
akashh@akashhh:~$ gedit sample.txt
akashh@akashhh:~$ hdfs dfs -put  ~/sample.txt /testing
```

```
akashh@akashhh: ~
                Bytes Written=241
23/06/29 21:40:39 INFO streaming.StreamJob: Output directory: /testing/output
akashh@akashhh:~$ hdfs dfs -ls -R /
drwxr-xr-x   - akashh supergroup          0 2023-06-29 21:40 /testing
drwxr-xr-x   - akashh supergroup          0 2023-06-29 21:40 /testing/output
-rw-r--r--   1 akashh supergroup          0 2023-06-29 21:40 /testing/output/_SU
CCESS
-rw-r--r--   1 akashh supergroup        241 2023-06-29 21:40 /testing/output/par
t-00000
-rw-r--r--   1 akashh supergroup        240 2023-06-29 21:40 /testing/sample.txt
akashh@akashhh:~$ hdfs dfs -cat /testing/output/part-00000
A       3
An      1
Because,        1
Elephant        1
Hadoop  2
Hadoop. 2
He      1
King!   1
Useful  1
an      1
and     2
at      1
core,   1
```

Browsing HDFS

localhost:50070/explorer.html#/testing/output

Hadoop Overview Datanodes Snapshot Startup Progress Utilities

Browse Directory

/testing/output Go!

Permission	Owner	Group	Size	Last Modified	Replication	Block Size	Name
-rw-r--r--	akashh	supergroup	0 B	29/6/2023, 9:40:38 pm	1	128 MB	_SUCCESS
-rw-r--r--	akashh	supergroup	241 B	29/6/2023, 9:40:38 pm	1	128 MB	part-00000

Hadoop, 2016.

akashh@akashhh: ~

```
        File Output Format Counters
                Bytes Written=12797
akashh@akashhh:~$ hdfs dfs -cat /output/part-r-00000
The Day is Cold Day :20200101   -21.8
The Day is Cold Day :20200102   -23.4
The Day is Cold Day :20200103   -25.4
The Day is Cold Day :20200104   -26.8
The Day is Cold Day :20200105   -28.8
The Day is Cold Day :20200106   -30.0
The Day is Cold Day :20200107   -31.4
The Day is Cold Day :20200108   -33.6
The Day is Cold Day :20200109   -26.6
The Day is Cold Day :20200110   -24.3
The Day is Cold Day :20200111   -25.0
The Day is Cold Day :20200112   -22.4
The Day is Cold Day :20200113   -21.9
The Day is Cold Day :20200114   -17.8
The Day is Cold Day :20200115   -20.8
The Day is Cold Day :20200116   -19.1
The Day is Cold Day :20200117   -21.1
The Day is Cold Day :20200118   -22.2
The Day is Cold Day :20200119   -22.8
The Day is Cold Day :20200120   -24.0
The Day is Cold Day :20200121   -26.3
```

localhost:50070/explorer.html#/output

Hadoop Overview Datanodes Snapshot Startup Progress Utilities

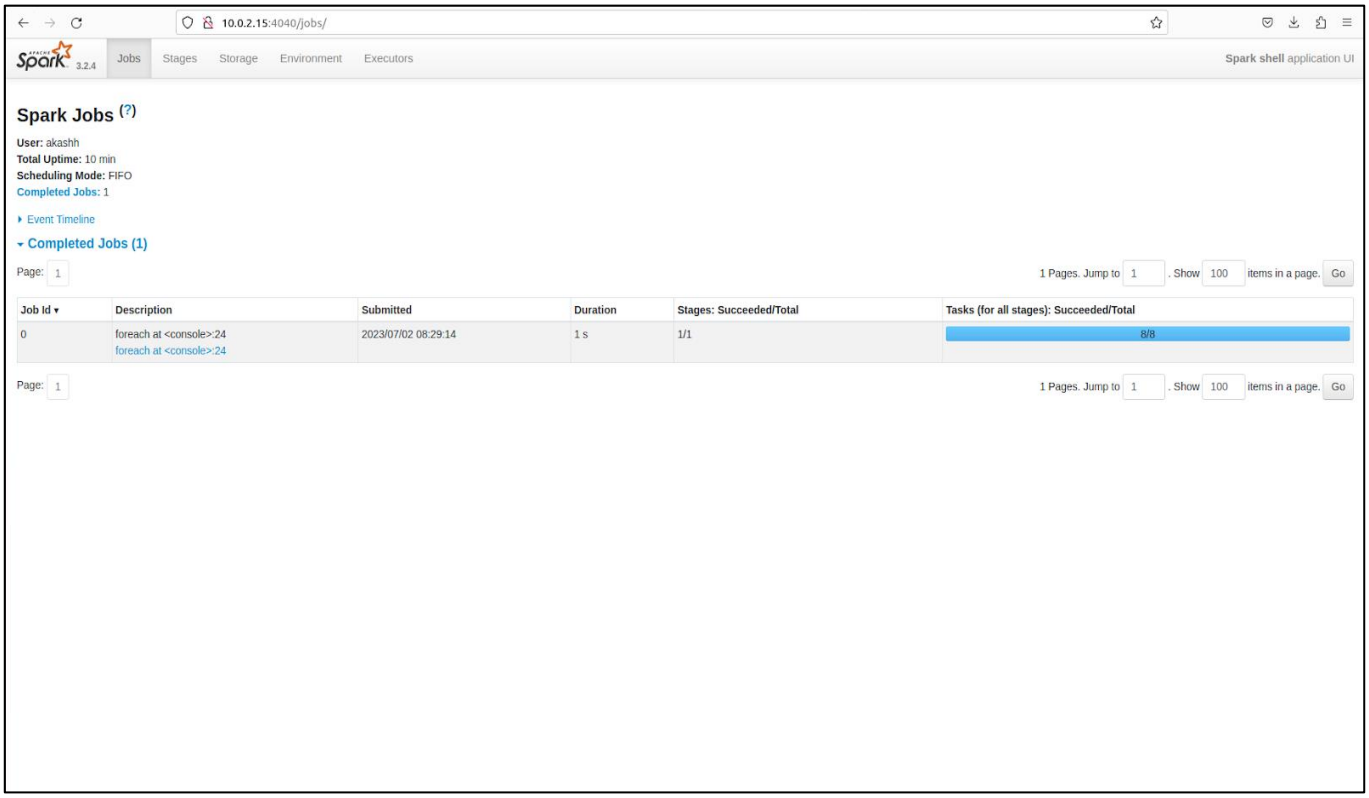
Browse Directory

/output Go!

Permission	Owner	Group	Size	Last Modified	Replication	Block Size	Name
-rw-r--r--	akashh	supergroup	0 B	2/7/2023, 10:01:27 am	1	128 MB	_SUCCESS
-rw-r--r--	akashh	supergroup	12.5 KB	2/7/2023, 10:01:27 am	1	128 MB	part-r-00000

Hadoop, 2016.

```
hadoop@mca-ThinkCentre-neo-50t-Gen-3: ~/Desktop
default
Time taken: 0.246 seconds, Fetched: 1 row(s)
hive> CREATE DATABASE IF NOT EXISTS Student_Details;
OK
Time taken: 0.049 seconds
hive> SHOW DATABASES;
OK
default
student_details
Time taken: 0.023 seconds, Fetched: 2 row(s)
hive> use Student_Details;
OK
Time taken: 0.028 seconds
hive> CREATE TABLE IF NOT EXISTS Student(Stud_id int, Stud_Name String, Stud_Ma
rk int, Stud_Dept String) ROW FORMAT DELIMITED FIELDS TERMINATED BY ',' LINES T
ERMINATED BY '\n' STORED AS TEXTFILE ;
OK
Time taken: 0.251 seconds
hive> ALTER TABLE Student RENAME TO Stud;
OK
Time taken: 0.126 seconds
hive> SHOW DATABASES;
OK
```

```
hadoop@mca-ThinkCentre-neo-50t-Gen-3: ~/Desktop
hive> LOAD DATA LOCAL INPATH '/home/hadoop/student_details.txt' OVERWRITE INTO
TABLE Stud;
Loading data to table student_details.stud
OK
Time taken: 0.666 seconds
hive> SELECT * FROM Stud;
OK
10001   Aazim   99      MCA
10002   Antony  98      MBA
10003   Akashh  35      MCA
10004   Manoj   45      MBA
10005   Prem    89      ME
Time taken: 0.576 seconds, Fetched: 5 row(s)
hive> SELECT * FROM Stud WHERE stud_mark<50;
OK
10003   Akashh  35      MCA
10004   Manoj   45      MBA
Time taken: 0.208 seconds, Fetched: 2 row(s)
hive> CREATE VIEW pass AS SELECT * FROM Stud WHERE stud_mark<50;
OK
Time taken: 0.211 seconds
hive> SELECT * FROM PASS;
OK
10003   Akashh  35      MCA
```

```
akashh@akashhh: ~
akashh@akashhh:~$ chmod -R 775  spark-3.2.4-bin-hadoop2.7
akashh@akashhh:~$ sudo nano ~/.bashrc
[sudo] password for akashh:
akashh@akashhh:~$ source ~/.bashrc
akashh@akashhh:~$ spark-shell
23/07/02 08:27:26 WARN Utils: Your hostname, akashhh resolves to a loopback addr
ess: 127.0.1.1; using 10.0.2.15 instead (on interface enp0s3)
23/07/02 08:27:26 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another
address
Using Spark's default log4j profile: org/apache/spark/log4j-defaults.properties
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLeve
l(newLevel).
23/07/02 08:27:32 WARN NativeCodeLoader: Unable to load native-hadoop library fo
r your platform... using builtin-java classes where applicable
Spark context Web UI available at http://10.0.2.15:4040
Spark context available as 'sc' (master = local[*], app id = local-1688266653426
).
Spark session available as 'spark'.
Welcome to
      ____              __
     / __/__  ___ _____/ /__
    _\ \/ _ \/ _ `/ __/  '_/
   /___/ .__/\_,_/_/ /_/\_\   version 3.2.4
```